MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION – October 2025

Third SEMESTER

25900359 - Discipline Specific Elective (DSE) - MG3DSEECT201 - INTRODUCTION TO MECHATRONICS

Duration: 2 Hours Maximum Marks: 70

Answer Key/Scheme with Questions

Part A – Multiple Choice Questions (1 mark each)

1. What is the role of resistors in R-2R ladder DAC?
   Answer: b) Form a voltage divider network
2. Which of the following real-world applications is best described as a mechatronic system?
   Answer: b) Electronic fuel injection system in cars
3. Define the primary goal of mechatronics in manufacturing?
   Answer: c) Enhance productivity and precision
4. Thermocouples work by:
   Answer: c) Temperature difference between metals
5. Sensor sensitivity is important because it:
   Answer: b) Determines smallest detectable change
6. Define the primary function of a hydraulic actuator?
   Answer: c) Convert fluid power into mechanical motion
7. Explain what is the common use of a relay in home automation?
   Answer: c) Controlling large appliances remotely
8. Proximity sensor that works on the principle of electromagnetic induction is
   Answer: c) Inductive
9. Identify which gear system is suitable, if a machine requires high-speed reduction?
   Answer: c) Worm Gear
10. Name the actuator that produces straight-line motion?
    Answer: d) Linear actuator
11. The main advantage of a closed-loop system is:
    Answer: b) Higher accuracy and automatic correction
12. SPI stands for:
    Answer: a) Serial Peripheral Interface
13. A system that changes when time is shifted is:
    Answer: b) Time-variant
14. A signal that exists only at specific time intervals is:
    Answer: b) Discrete signal
15. ZigBee is mainly used for:
    Answer: a) Short-range, low-power communication

16. Why is Python often preferred in rapid prototyping?
    Answer: a) Easy syntax and rich libraries
17. Which is NOT an actuator?
    Answer: d) Potentiometer
18. Why are collaborative robots (cobots) important in modern industries?
    Answer: b) They work safely alongside humans
19. The Arduino function to read analog input is
    Answer: c) analogRead()
20. Which control system ensures better robotic arm accuracy?
    Answer: c) PID

Part B – Short Answer Questions (5 marks each)

21. Explain why a SAR ADC is commonly used in mechatronics.

Explain at least 2 points. 2.5 marks each

The Successive Approximation Register (SAR) ADC is one of the most widely used analog-to-digital converters in mechatronic applications because it provides an excellent balance between speed, accuracy, and power consumption. The SAR ADC converts an analog signal into a digital form by comparing the input voltage with successively generated reference voltages using a binary search algorithm. This method enables moderate-to-high conversion speed with high resolution, typically between 8 to 16 bits.

In mechatronic systems, real-time data acquisition is critical—for example, when sensors continuously monitor position, temperature, or pressure. The SAR ADC's fast conversion time allows the microcontroller to quickly process signals and generate control actions. Moreover, its relatively low power consumption and compact size make it ideal for embedded mechatronic devices, such as robotic systems, motor controllers, and instrumentation modules.

22. Describe the importance of controllers in a mechatronic system.

Explain the role of controllers in detail – 5 marks

Controllers are the "brains" of a mechatronic system. They process data received from sensors, compare it with the desired set point, and generate suitable control signals to drive actuators. This ensures that the system performs its intended function accurately, efficiently, and safely. Common controllers used in mechatronic systems include microcontrollers, PLCs (Programmable Logic Controllers), and PID (Proportional-Integral-Derivative) control units.

For example, in an automatic robotic arm, the controller continuously receives feedback from position sensors and adjusts motor movements to achieve precise motion control. Controllers also handle decision-making, timing, and safety monitoring. Without controllers, the interaction between mechanical and electronic components in a mechatronic system would be uncoordinated and unreliable, leading to poor performance or system instability.

23. Identify the medium used in pneumatic systems and explain.

Medium identification – 1 mark. Explanation – 4 marks

The working medium in pneumatic systems is **compressed air**. Air is readily available, non-polluting, and safe to use, which makes it an ideal choice for industrial automation and mechatronic systems. Pneumatics relies on the energy stored in compressed air to produce linear or rotary motion in actuators such as cylinders and motors.
Compressed air is easy to transport through pipelines and can be stored in reservoirs for continuous operation. Pneumatic systems are commonly used in manufacturing lines, robotics, and packaging machines where quick, repetitive motion is required. Additionally, the system components are relatively simple and robust, which makes pneumatic systems reliable and cost-effective for automation applications.

24. Show how would you use a potentiometer to measure angular position.

Explanation the working of a potentiometer and how it measures the angular position – 5 marks

A potentiometer can be used as an **angular position sensor** because its resistance varies linearly with the rotation of its shaft. When a constant voltage is applied across the potentiometer terminals, the output voltage taken from the wiper terminal changes proportionally to the angular displacement of the shaft. This forms a simple voltage divider circuit that provides an analog voltage corresponding to the angle of rotation.
By interfacing the output voltage to an analog-to-digital converter or microcontroller, the system can accurately determine the angular position. This method is simple, low-cost, and widely used in applications such as joystick controls, robotic arms, and servo motor feedback.

25. What is the role of an anti-aliasing filter?

Explain the working of anti-aliasing filter – 5 marks

An anti-aliasing filter is a low-pass filter placed before an analog-to-digital converter

(ADC) in signal processing systems. Its main function is to remove or attenuate high-frequency components of an analog signal that are above half the sampling frequency (known as the Nyquist frequency). If these frequencies are not filtered out, they can "fold back" into the lower frequency range after sampling, creating distortion called aliasing.

In mechatronic systems, signals from sensors often contain unwanted high-frequency noise caused by environmental factors or electrical interference. The anti-aliasing filter ensures that only the relevant frequency components of the signal are converted into digital form, preserving signal integrity.

26. Define attenuation in signal processing.

Definition of attenuation – 1 mark, How it occurs and how it can be avoided – 4 marks

Attenuation refers to the **reduction in amplitude or power** of a signal as it travels through a transmission medium or an electronic circuit. It is often expressed in decibels (dB) and can result from resistance in conductors, dielectric losses, or intentional filtering. In communication and control systems, attenuation determines how much signal strength is lost between the source and the receiver.

In mechatronic and instrumentation applications, attenuation may occur in sensor cables, amplifiers, or filters. For example, long cable runs in industrial systems can cause voltage drops, affecting sensor readings. Understanding attenuation helps engineers design compensation circuits or amplification stages to maintain signal accuracy and system reliability.

27. Explain how the proportional term affects system performance.

Explain the role od proportional term in PID controller – 5 marks

In a proportional control system, the output control signal is directly proportional to the **error**—the difference between the desired set point and the actual process value. The proportional term helps the system respond quickly to changes, thereby reducing large errors. Increasing the proportional gain improves response speed but may also introduce overshoot or oscillations if set too high.

The proportional control contributes to system stability and accuracy when combined with other control terms, such as integral and derivative (in a PID controller). However, proportional control alone cannot eliminate steady-state error completely. It is often used where small errors are acceptable, or as part of a more advanced PID control strategy to fine-tune system performance.

28. Explain why Python is often used in embedded system prototyping.

Describe at least two reasons in detail – 2.5 marks each

Python has become a popular language for **rapid prototyping in embedded systems** because of its simplicity, readability, and vast library support. It allows engineers to develop and test algorithms quickly before implementing them in lower-level languages like C or C++. Libraries enable direct hardware interaction with devices like Raspberry Pi, Arduino, and ESP32.

Python also supports quick data visualization and debugging, which makes it ideal for testing control algorithms, sensor data acquisition, and IoT applications. Its cross-platform nature and integration with machine learning and cloud tools further extend its use in modern mechatronic systems. Thus, Python bridges the gap between concept design and practical implementation, speeding up the overall development process.

Part C – Essay Questions (10 marks each)

29. Explain Mechatronics with suitable application.

Explain mechatronics and its components – 5 marks, Any one application -5 marks

Mechatronics is an interdisciplinary field combining **mechanical engineering, electronics, computer control, and systems design** to develop intelligent and automated machines. Instead of treating these areas separately, mechatronics integrates them to achieve higher **precision, reliability, and flexibility** in modern systems such as robots, automobiles, and smart devices.

30. Use an appropriate actuator (DC, stepper, or servo motor) for a robotic arm and justify your choice.

Servo motor, its working and advantages – 10 marks

In a robotic arm system, actuators are the components that provide the required **motion and force** for positioning and manipulating objects. The choice of actuator depends on parameters such as **accuracy, torque, speed, and control requirements**. Among the commonly used actuators—**DC motors, stepper motors, and servo motors**—the **servo motor** is the most suitable for a robotic arm application.

31. What are the typical applications of FIR filters in signal processing?

Define IIR filter – 2 marks. Applications – 8 marks

A **Finite Impulse Response (FIR) filter** is a type of digital filter whose response to an impulse settles to zero in finite time. It is widely used in signal processing due to its **stability, linear phase characteristics**, and ease of implementation in both hardware and software. Unlike IIR (Infinite Impulse Response) filters, FIR filters do not have feedback loops, ensuring inherent system stability.

32. List and explain different types of proximity sensors used in automation.

Explain at least 4 types. 2.5 marks each

A **proximity sensor** is a device used to detect the presence or absence of an object without making physical contact. It operates by emitting and receiving energy fields such as electromagnetic, optical, or ultrasonic waves. These sensors are essential in **automation, robotics, and industrial control systems**, where they provide reliable object detection, positioning, and safety interlocks.

Types of Proximity Sensors and Their Working Principles:

1. Inductive Proximity Sensor:
2. Capacitive Proximity Sensor:
3. Ultrasonic Proximity Sensor:
4. Infrared (IR) Proximity Sensor: